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RESEARCH-ARTICLE

Understanding Skill Transfer Between University and Workplace Through Reflective Practice: A Software Engineering Work-Based Learning Experience

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Abstract

Work-based learning (WBL) programs in Software Engineering provide students with opportunities to apply academic knowledge in the workplace. However, how students navigate the transition between knowledge and skills acquired at university and their application in the workplace remains largely unexplored. We implemented a WBL Software Engineering program in which students spend most of their time in the workplace rather than at university. Reflective student essays were used to examine how our implementation supports the integration and transfer of learning between academic and professional contexts. University modules – such as those on software design patterns and programming – offered directly applicable skills, while foundational knowledge and problem-solving frameworks supported further development. Workplace experiences reinforced understanding, provided context, and enabled practical application. However, skills such as project management, leadership, and context switching were primarily developed in the workplace, revealing gaps between academic team projects and industry needs. Difficulties in transferring some advanced technical skills also point to a disconnect between academic instruction and industry application. Our findings suggest a cyclical relationship between university and workplace learning, with each enhancing the other. We propose key considerations for WBL program design, including curriculum alignment, structured skill development, and integrated delivery. Finally, we underscore the value of reflective essays in observing and supporting skill integration.

CCS Concepts

• **Social and professional topics** → **Computing education**;
Software engineering education; *Computing profession*.

Keywords

work-based learning, reflective practice, software engineering

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1 Introduction

Work-based learning (WBL) programs in Software Engineering education provide an alternative pathway for students, widening access by combining academic studies with professional experience in the workplace. These programs are designed to address the challenges of bridging the gap between the theoretical knowledge taught at the university and the practical skills demanded in industry. Internationally, WBL models, such as cooperative education (co-op) programs in the United States, demonstrate the potential to integrate academic and workplace experiences to enhance professional and technical skills, providing valuable context for examining WBL in software engineering education [6]. Similarly, studies have highlighted how WBL programs foster critical employability skills, such as communication, adaptability, and self-management, through real-world applications and challenges [2].

The literature underscores the complementary roles of university education and workplace experiences in skill development. Universities are primarily responsible for equipping students with foundational technical skills such as programming, system design, and theoretical knowledge [15, 19, 21]. However, efforts to foster transferable skills such as communication and teamwork often lack consistency, with the depth of these skills' development not meeting industry expectations [19]. In contrast, workplace experiences provide essential opportunities for students to apply technical knowledge and refine professional dispositions, such as leadership, adaptability, and decision making, which are often underdeveloped in academic settings [13, 14]. The competency-based frameworks



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used in WBL programs have also been shown to support the assessment and development of these professional competencies by combining academic knowledge with practical skills and dispositions to meet industry needs [1].

A recurring theme in the literature is the gap between academic preparation and workplace demands. Universities aim to develop technical and cognitive skills, but critical professional dispositions are often overlooked [17]. Although frameworks such as the Skills Framework for the Information Age (SFIA) [16] help align academic programs with industry needs [8], competencies related to professional behaviour and leadership are best developed through direct workplace experience. Research also highlights the importance of transferable skills (those that can be applied in academic and professional contexts), yet note that these skills are often more effectively refined in the workplace [12].

Traditionally, skill transfer in education has been viewed as a one-way, or sequential, process, where foundational technical skills and dispositions learned at university are applied to workplace tasks. However, in WBL programs, a more dynamic feedback loop is observed, where workplace experiences influence students' academic perspectives and performance. Reflective practices, such as writing essays, play a critical role in this process by encouraging students to articulate the insights gained in the workplace and connect them to their academic studies [3, 7, 15]. This feedback loop not only enriches the students' learning experience, but also has the potential to inform curriculum design and teaching methods, hence aligning university programs more closely with industry needs.

In this experience report, we use reflective essays from students – an assessment component in their workplace-based modules – as a lens to investigate the research question *"How do Software Engineering students perceive and articulate the integration and transfer of skills between university education and workplace experiences in a WBL program?"*. These essays offer valuable insight into how students perceive and articulate skill integration and transfer in a WBL program, while also highlighting the role of reflective practice in making learning visible. By analysing them in the context of our program's design and implementation [20], we present practical insights, lessons learned, and recommendations to support skill development. This report provides actionable guidance for educators and program designers aiming to implement or enhance effective WBL programs in computing education.

The WBL program at the University of Glasgow, UK, is a four-year apprenticeship degree in software engineering developed in collaboration with industry [4, 20]. An apprentice student is recruited by an employer and enrolled in the program, where they will balance two blocks of intensive teaching with about 80% of the time spent in their work roles. Of particular relevance are two modules, each accounting for half of the total credits in an academic year, that apprentices take in their third and fourth years. The module specification pages for these modules are available at the following links: Workplace Assessment Year 3 and Workplace Assessment Year 4. These two modules have three associated assignments: a workplace-based project under the supervision of an academic and workplace line manager (contributing 60% to the module grade); a portfolio of work aligned with the UK-SPEC competency framework [11] (30%); and a reflective essay that encourages personal reflections on their journey to build those competencies (10%).

These essays are based on reflective workplace journals that students are encouraged to maintain throughout each of their final two years. Instructions require that students make specific references to the competency framework throughout their prose and make suggestions of topics students could consider addressing, such as examples of knowledge acquired at university and then applied at the workplace, or skills acquired solely or largely based on workplace experience. This collection of submitted essays forms the basis for the study in this work.

In this paper, the terms *module* and *course*, as well as *reflective reports* and *reflective essays*, are used interchangeably, reflecting student language and common terminological variation. For consistency, we primarily use *module* and *reflective essays* throughout.

2 Observations from Reflective Essays

This section outlines the methodology used to analyse students' reflective essays, presents the results of the analysis, and follows with a discussion of the findings.

2.1 Data Collection and Analysis

2.1.1 Data Collection. As part of the assessment for each of the work-based year-long modules in the last two years of the program, students are required to submit a 1500-2000 word reflective essay based on their portfolio of work. This was designed to encourage critical reflection of the student's personal journey in developing the competencies outlined in the competency framework designed for each module. Students are also encouraged to reflect on competencies beyond the framework based on individual experiences. The essays were assessed on: (i) quality of reflection (80%): depth and clarity of reflection, critical engagement with a wider perspective (e.g. peer interaction or literature) and specific references to competencies; (ii) quality of academic communication (20%): clarity, structure, and professionalism in writing. In addition, students were guided to maintain reflective workplace journals to aid in the writing process, helping them capture ongoing insights. This assessment emphasises high-level reflection, categorised as reflection-on-action, which goes beyond immediate reactions to involve longer timeframes and engagement with others and theoretical ideas. To assist students, several guiding topics were suggested, including: competencies gained implicitly and how they were achieved; comparison between knowledge and skills acquired at university versus the workplace; applications of university-acquired knowledge and skills in real-world settings; skills and professional dispositions developed solely or primarily in the workplace; broader reflections on personal and professional growth.

We collected data from 58 students enrolled in our WBL program in software engineering during the academic year 2023-2024. The cohort consisted of 21 third- and 37 fourth-year undergraduate students employed in software engineering roles that can relate to different aspects of the software engineering life cycle in 13 companies in the tech sector, including start-ups, large corporations, and public sector organisations. At the time the assessments were submitted, the median age for Year 3 students was 20 years with mean 22.7, while for Year 4 the median was 24 and mean 25.7.

Ethical approval to use the essays was obtained from the Ethics Committee of the College of Science and Engineering, University

of Glasgow (approval number: 300230060, Dec. 1st, 2023) as part of a broader program evaluation. To ensure anonymity, all student names were replaced with pseudonyms while preserving their demographic characteristics to maintain the integrity of the analysis.

2.1.2 Data Analysis. A form of thematic analysis was used to make sense of the student reflections that form the basis of this experience report. A sample of the data (representing approximately 50% of the reflective essays of Year 4 students) was analysed using an inductive thematic approach to identify common themes. These themes provided a framework for analysing the complete data set using a deductive approach. As such, our analysis followed many of the tenets of thematic analysis, as set out by Braun and Clarke [10], but referred to the inductive approach described by Boyatzis [9] to guide our development of the themes.

We now present the results of our analysis, first focusing on skills acquired at university that students found useful in the workplace, followed by skills developed predominantly through workplace experiences. Each subsection corresponds to a broad category, within which the themes identified through deductive thematic analysis are presented as headings (see Sections 2.2.1-2.2.6 and 2.3.1-2.3.5, respectively), each theme highlighting specific skills evidenced through students' reflections. We conclude each subsection with a discussion of contradictions and disagreements.

2.2 Skills acquired at university that are found useful in the workplace

2.2.1 Software Development and Engineering Practices. The evidence from students' reflective essays demonstrates how foundational software development and engineering practices taught at university provide the groundwork that is then refined and contextualised in the workplace. This exemplifies the sequential nature of the development of skill in apprentices.

For example, Mark Simpson's reflection on the module *Software Product Release Engineering (SPRE)* illustrates the direct alignment between university curriculum and industry practices: "*SPRE has taught me verbally and practically the stages involved in preparing and deploying a release.*" This is reinforced by Mohammed Rizwan's observation about learning "*the often-neglected practice of SPRE – the process in which software is promoted and deployed into a production environment, so that it can be used by end users*". This shows how specific technical coursework prepares students for real-world deployment scenarios.

The application of design patterns provides another example of successful curriculum alignment with industry needs. John Smith's Year 3 reflection demonstrates how theoretical knowledge transforms into practical workplace value: "*I have been able to apply my knowledge of design patterns to create solutions for complex problems in an efficient and maintainable way*". Importantly, he notes that "*these design patterns are standard across the industry*", highlighting how university teaching of fundamental concepts directly supports workplace communication and collaboration.

Thomas Day's experience with lambda calculus, meanwhile, illustrates the progression from academic theory to workplace application: "*Using lambda calculus in recent projects was difficult to implement at first. However, once accustomed to the syntax and small*

logic changes, I am now capable of comfortably including basic elements of the theory in this setting". This narrative suggests that the technical fundamentals that are introduced in university are then refined through workplace experience.

These examples suggest that curriculum design should prioritise alignment between foundational technical modules and industry practices, while recognising that practical implementation skills can develop further in the workplace setting. The students' experiences seem to support our assumption that technical skills develop best in sequence – from theoretical understanding in university to practical application and refinement in the workplace.

2.2.2 Teamwork and Project Management. Student reflections on teamwork and project management highlight both the value and limitations of university preparation for workplace collaboration. Here, the evidence suggests that while university can initiate the development of these skills, significant aspects of project management and leadership are best developed in the workplace context.

Scott Anderson's Year 3 reflection demonstrates how university group work does provide foundational teamwork experience: "*At university I have taken multiple courses which have required a lot of group work, the working being completed ranging from programming to presentations. From these classes I have learned how to work in a team and plan resources*". This illustrates the basic transferable skills developed in the academic setting, although notably focuses on fundamental rather than advanced team dynamics. The progression to more complex team management skills is also evidenced in Ivor Pie's Year 3 reflection: "*The course also covers interpersonal and management skills that are needed for mentoring and leading teams of software developers. The course focuses on whole-team coaching and how to introduce and embed software engineering practices in existing teams*". So, it might be argued that it is useful to build leadership preparation into a WBL curriculum, while acknowledging that these skills will be developed primarily in workplace settings.

Fraser Aitken's Year 4 reflection on the taught *Advanced Professional Software Engineering* module demonstrates how university courses can introduce industry-standard project management tools: "*This involved me working as a team to understand agile concepts and the importance of using task tracking platforms such as Jira*". This reinforces the idea that we should ensure that our technical courses align with industry practices, particularly in a WBL software engineering program.

Colin Barclay's Year 4 reflection on coaching experience illustrates the development of transferable leadership skills: "*The university coaching experience has not only equipped me with transferable leadership skills but has also deepened my capacity for critical thinking, problem solving, and adaptability*". Again, this example demonstrates the value of university preparation while highlighting how these skills are connected to workplace demands.

Based on these student experiences, it appears that the university can provide important groundwork in teamwork and project management, but institutions should recognise that many leadership and management skills will develop primarily through workplace experience. Evidence suggests that university programs should focus on providing foundational collaborative experiences while preparing students for the more complex team dynamics they will encounter in the workplace.

2.2.3 Communication and Presentation Skills. Reflections on communication and presentation skills also suggest that these skills benefit from initial development in university before being adapted to workplace contexts. Anthony Allan’s Year 4 reflection demonstrates the concrete value of university presentation requirements: *“This year alone in university I have had to make four different presentations which contribute to my final grade. I was able to secure an A grade in the two which I have marks back for which lets me know that I am improving on a public speaking front.”* This structured development of presentation skills, with formal feedback, provides a foundation that students can later adapt to workplace needs.

Meanwhile, Jack Archibald’s Year 3 reflection indicates how university presentations build versatile communication capabilities: *“One of the skills I have most developed throughout my university learning has been my ability to create and deliver presentations, across many different knowledge domains, which are engaging, informative, and which keep to time”.* This supports our conclusion about the transferability of communication skills from academic to workplace settings. David Hook’s Year 3 reflection particularly illustrates the bridging between academic and workplace communication contexts: *“The presentation projects at University also gave me a clear understanding of various ways to present information in a concise and clear format within a time limit. This was useful, as while there is no strict time limit when presenting at work, it allowed me to practice keeping the audience engaged and be concise with my main points.”*

These examples validate our program design approach of building in presentation and communication components, as they clearly transfer well to workplace settings. This is an important realization for a student enrolled on any kind of program, not just a WBL degree: the presentations they are required to give at university really *do* prepare them for industry, despite students’ well-documented anxieties around public speaking [18]. The examples here also demonstrate the potential value of our reflective assignment approach, as students are able to articulate how they adapt their academic communication skills to workplace requirements.

The evidence suggests that formal university training in communication provides essential structure and feedback that help students develop confidence and fundamental techniques, which they can then modify for different professional contexts. This is another example of where the sequential nature of skill development in a WBL program is important to consider while highlighting communication as an area where university preparation directly enhances workplace performance and confidence.

2.2.4 Research and Analytical Skills. Student reflections on research and analytical skills suggest that problem-solving frameworks taught in university can also be effectively applied and expanded in workplace contexts. The data also highlight the value of reflective assignments in helping students connect academic and workplace learning.

Malak Ahmed’s Year 3 reflection directly demonstrates how research methods training transfers to workplace needs: *“One of the most important skills that I have worked upon in university that I have found useful in the workplace was from a recent class I took Research Methods and Techniques... Through the class I learned the proper ways of searching for and reviewing research papers and reviewing them properly.”*

So, while a module on research methods and techniques may not immediately seem relevant to the development of an apprentice software engineer, the module in question appears to provide another example of how foundational course material can align with industry practices. Michael Jones’s Year 4 reflection also refers to this module, and highlights the bidirectional nature of skill development between university and workplace: *“This previous year, the Research Methods and Techniques module helped hone my research skills, which covered various techniques used by academics; however, over the last four years as a developer, exploring concepts and technologies has been an integral part of dealing with issues.”*

Such examples suggest how workplace experiences can inform academic understanding while allowing workplace challenges to drive further academic inquiry.

These reflections support our program design principles [5, 20] in two ways. First, they demonstrate how academic research skills provide frameworks that students can adapt to workplace problem solving. Second, they show how workplace experiences help students appreciate and apply academic research methods in practical contexts. Our students’ experience suggests that research and analytical skills represent an area where university preparation provides crucial foundational approaches that students can refine through workplace application.

2.2.5 Programming and Technical Skills. The reflections on programming and technical skills provide evidence that technical fundamentals are best developed sequentially – introduced in university, then enhanced through workplace application. Nadine Stevens’s Year 4 reflection neatly illustrates this progression: *“My education started with university with basic concepts like SQL queries and key systems, and then got refined through my work with various stored procedures, views, and queries”.*

David Hook’s Year 3 reflection provides strong evidence for how university technical preparation enables workplace success: *“University courses such as How to Learn a New Language, were pivotal in teaching me multiple ways to approach development, assisting with mastering the basics of any language, which in turn helped me when it came to JavaScript and React development in the workplace and allowed me to extend my approach to development as a software engineer.”*

This highlights the importance of aligning university curricula with industry needs while teaching adaptable technical foundations. Callum Blair’s Year 3 comment about proof-of-concept development further illustrates how technical knowledge transfers to significant workplace projects and real-world challenges: *“The insights gained from the proof of concept development phase were invaluable in ensuring that our migration to AWS was successful”.*

These reflections arguably support our approach to curriculum design, demonstrating that foundational programming and technical modules need to directly align with industry practices while providing students with adaptable skills they can apply to various workplace technologies.

2.2.6 Problem-solving and Critical Thinking. Students’ reflections on problem solving and critical thinking demonstrate how university-taught frameworks provide a foundation that students later apply and expand upon to solve problems in the workplace. Jennifer Johnson’s Year 3 reflection, for example, provides a broad perspective

on problem-solving skill development: “*University has been beneficial for developing general problem-solving skills and learning my general areas of interest and strength*”. This suggests perhaps that the foundational frameworks provided by universities are more general than specific but are valuable nonetheless.

Elizabeth Jack’s Year 4 comment on the aforementioned *Research Methods and Techniques* module – “*has enabled me to critically evaluate systems*” – further demonstrates how academic training provides analytical tools that transfer to workplace contexts. Meanwhile, Amber Fellows’s Year 4 reflection illustrates how academic skills transfer to workplace needs: “*I developed the skill of being able to comprehend complex information and communicate it effectively in my own writing. The course also taught me how to properly reference sources, which has been a valuable skill in my current work*”. This example again highlights the sequential nature of skill development, showing how academic foundations are adapted to workplace requirements.

2.2.7 Discussion. Our analysis of the students’ reflections on the transfer of knowledge and skills from university to the workplace reveals some contradictions and disagreements. Indeed, while many students found their university education directly applicable to their work, some experienced a disconnect. For example, Tony Lucas noted: “*Throughout all of university we are taught to develop in IDEs and up to date tools to make our programming lives easier and within these IDEs there are insightful error codes. In my role I do not have access to these tools or insightful error codes, instead I write my code on Linux command line with error codes which could mean anything making programming a lot harder.*”

Some students valued the theoretical foundation provided by the university, while others found the workplace experience more valuable. Clara Bond, for example, emphasised the complementary nature: “*Although university provides me with the building blocks, I think a lot of the knowledge required for solutions needs to be picked up in industry for your specific context*”. In contrast, Arnold Tusk noted that some of the theory picked up at university proved unexpectedly useful in a real-world context: “*I found it surprising that I was able to so readily think of practical applications of the Usability Heuristics, as during university, I had found it fairly challenging to identify instances of those heuristics being applied on existing web pages*”.

2.3 Skills acquired solely or largely based on workplace experience

2.3.1 Project Management and Leadership. Reflections on project management and leadership emphasise that some skills will develop primarily in the workplace. Fred Jackman’s Year 4 reflection directly supports this finding: “*The most prominent skill that I have developed this year within the workplace were my project management skills as I believed them to one of the most vital abilities to develop in the workplace*”.

Meanwhile, Jeff Allison’s Year 3 reflection provides insight into how leadership skills develop through workplace responsibility: “*During this feature I was taking ownership of my own backlog of tickets and deciding what was required to implement tickets and defining acceptance criteria to contribute to the delivery of the overall feature*”. This example highlights the reality of workplace versus

academic projects, showing how workplace contexts provide unique opportunities to develop project management capabilities.

These reflections support our program design recommendations in two ways. First, they confirm that certain skills, particularly in project management and leadership, develop most effectively through workplace experience. Second, they suggest that, while university programs might introduce these concepts, real development occurs through practical application in professional settings.

2.3.2 Communication and Interpersonal Skills. Although we note that communication skills development can and should be initiated at university, the reflections here reveal how workplace contexts provide distinct opportunities for developing more nuanced interpersonal abilities, building on those university foundations. Jamie Bradshaw’s Year 3 reflection, for example, illustrates how workplace communication skills develop through iterative feedback in ‘real’ contexts: “*The main way my presentation skills have developed are by having smaller demonstrations with my colleagues who I have gotten to know over the years and asking for feedback by the end, i.e., asking if I have explained that clearly, if I went too fast, or if there is anything I could do better, which I can then apply to my next presentations.*”

Colin Barclay’s Year 4 reflection implies a progression to more sophisticated communication: “*Daily meetings became a platform where I could confidently share updates, troubleshoot issues, and articulate my vision for various implementations*”. Meanwhile, Anthony Allan’s Year 4 reflection further demonstrates how workplace environments foster broader communication capabilities: “*Within my work life, I am consistently speaking to people from all over the world and even senior employees in a relaxed and confident manner*”. As noted above, this demonstrates that we should accept that in a work-based program certain aspects of professional skills will develop primarily in the workplace.

2.3.3 Technical Skills Specific to the Work Environment. Some student reflections refer to the development skills they have learned exclusively in the workplace, particularly those related to more advanced technical concepts. Tony Lucas, for example, notes: “*I remember a year ago I tried using lambdas in Python for something and I could not get it to work or fully understand how they worked but I am now using them in language which I am not as strong in and incorporating them in a codebase which is not my own.*”

Meanwhile, Archie Hamilton goes so far as to suggest that the material taught actually hindered his understanding of a particular concept: “*One such technique is the use of metaclasses, which I believe are inherently difficult to understand due to the way in which introductory courses (such as How To Learn a New Language) introduce classes*”.

2.3.4 Problem-solving and Critical Thinking. Jack Archibald’s Year 3 reflection illustrates the different stakes and complexity of workplace problem solving compared to academic projects: “*When I came to be working on the [proof of concept] in question, I found that the majority of the work to be done was not in the domain of implementation, but rather was about design; Before we could begin implementation, we needed to decide what kind of approach would be suitable, and making the wrong choice could mean weeks or months of work wasted – which was never a risk on my university projects.*”

Meanwhile, Clara Bond’s Year 4 reflection demonstrates how workplace problems drive deeper learning: *“Problem-solving in complex environments was notably enhanced through identifying and resolving an API issue that became a critical learning moment”*. Bjorn Ekman’s reflection shows how workplace problem-solving develops collaborative approaches: *“This experience led me to also adopt a similar approach in my problem-solving approach which allowed me to combat blockers more efficiently by working with others to solve common challenges”*. These reflections relate our program design principles about the sequential development of problem-solving skills, from academic frameworks to workplace application. However, they also alert us to the need to prepare students for the different implications of workplace versus academic projects.

2.3.5 Adaptability and Continuous Learning. These reflections illuminate why some competencies cannot be fully simulated in university environments. Take, for example, Ben Ratcliffe’s Year 4 reflection: *“One of the many skills I have developed solely at the workplace is the ability to quickly context switch between different streams of work”*. Certainly, it seems reasonable to expect that context-switching abilities are best developed through workplace experience rather than academic preparation.

Mark Simpson’s Year 4 reflection further demonstrates how workplace realities develop practical adaptability: *“My ability to be realistic and accept when adjustments must be made in the face of limited time and resources has certainly improved”*. Again, this illustrates that certain skills will develop primarily in the workplace context.

Callum Blair’s Year 3 reflection similarly illustrates how workplace challenges foster adaptability: *“An implicit competency I gained without deliberately planning for was adaptability and crisis management. The large AWS migration efforts tested my ability to adapt and manage unexpected challenges”*. So, while we can better prepare students for the realities of work, some skills inherently require the authentic context of professional practice.

2.3.6 Discussion. Again, there is some disagreement and nuance to be found in the students’ reflections on the skills acquired in the workplace. For example, some students find the knowledge in the workplace to be deeper and more specialised, while others view the knowledge in the university as more comprehensive. Luca Stewart suggests that the depth of knowledge required at university is limited, *“compared to a multinational company with hundreds of different data sources”*. However, Alex Findlay indicates that workplace knowledge can be shallow: *“All knowledge [in the workplace] is treated on a need to know at that moment basis”*.

As noted above, students appreciate the theoretical foundation from university, while others find it less useful compared to practical workplace experience. Clara Bond values both, echoing the idea that foundational skills and knowledge gained at university are further developed in the workplace: *“Theoretical knowledge gained from undergraduate courses provided a platform for me to tackle workplace difficulties with a critical, analytical perspective. However, applying this knowledge in a practical setting cemented my understanding and skill set.”*

However, Bjorn Ekman seems to value practical knowledge more highly, although acknowledging that the topics covered at university are interesting: *“I feel these knowledge points do not transfer well*

into the practical working environment”. Indeed, workplace skills are often learned in context and with real-world consequences, while university skills are learned in a more controlled environment. Martha Mu reflects: *“Workplace learning, and development has allowed me to fully understand and apply this in a practical setting. Learning skills such as SQL and relational database content in the context of a closed coursework was not as helpful as hands on, on the job practice.”* However, Arnold Tusk offers a different perspective: *“In the workplace, as part of my personal project, I needed to apply these requirements gathering techniques to produce a clear specification of my project that I could propose to my team. The skills that I acquired at university proved to be invaluable.”*

3 Conclusions and Future Work

Our experience of investigating the implementation of a work-based Software Engineering program based on analysing students’ reflective learning reports suggests a number of factors for colleagues to consider in the design of WBL. To facilitate the integration and transfer of learning between university education and workplace experiences, perhaps the first aspect to consider is *curriculum design*. Specifically, it is important to ensure that the foundational technical modules (such as programming, design patterns, and software release engineering) align directly with industry practices. We should also (continue to) build in presentation and communication components as these transfer well to workplace settings. We might also consider adding more explicit project management and leadership preparation, although it can be argued that these skills are best learned in the workplace.

With regard to *skill development structure*, we should recognise that some skills develop best in sequence. For example: technical fundamentals covered at university first, then refined in the workplace; communication skills started in university, then adapted to workplace contexts; and, problem-solving frameworks taught in university, then applied, and expanded in workplace. We also suggest that we accept that some skills, such as context switching, adaptability, and advanced technical implementation, will develop primarily in the workplace.

In terms of how to ensure *program integration*, we advocate for the use of reflective assignments to help students connect university and workplace learning. Assessing student learning in this way provides a mechanism for workplace experiences to inform academic understanding, while allowing workplace challenges to inspire further academic inquiry. Future work could investigate how reflective essays support students in internalising their learning and engaging in higher-level reflections on skills and competencies. Assignments should look for evidence of both technical and professional skill development, and assess how well students adapt academic knowledge to workplace contexts. Colleagues might also wish to consider both junior and senior perspectives (years 3 and 4, in our case) to understand skill progression. Finally, learning on the job is at the heart of WBL, and students should be expected to develop different skills in the university and workplace contexts. However, we can perhaps better prepare them for the reality of workplace versus academic projects. Further research could also explore long-term professional outcomes and refine curriculum alignment to enhance skill integration.

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